

Active learning for data-scarce multi-objective polymer design: Robust strategies with experimental validation

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ABSTRACT

Data-driven polymer design is fundamentally limited by severe experimental data scarcity and intrinsic trade-offs between competing material properties, such as mechanical stiffness and network mobility. While active learning (AL) offers a promising pathway for navigating high-dimensional design spaces, its practical deployment in experimental polymer systems remains hindered by the lack of robust strategies under strict data constraints. In this work, we establish an experimentally grounded active learning framework for data-efficient multi-objective polymer optimization. By systematically evaluating combinations of structured initial sampling and acquisition strategies within a Gaussian Process framework, we identify that the coupling between initialization and acquisition governs optimization efficiency under data scarcity. The framework is experimentally validated on a polyester-based vitrimer system with a highly combinatorial formulation space. With only 28 samples, including 12 initial points and 4 active-learning iterations, the approach rapidly expands the stiffness-mobility Pareto frontier, identifying non-obvious formulations that achieve a 34.5% reduction in stress-relaxation time and a 3.3% increase in Young's modulus. These results demonstrate that well-designed active learning strategies can extract meaningful Pareto improvements from minimal experimental data and provide practical guidance to accelerate polymer discovery under severe data scarcity. The present validation spans two systems with relatively continuous formulation–property landscapes, a literature-derived polyurethane benchmark, and the polyester-based vitrimer system; extension to highly constrained or high-dimensional design spaces remains to be established.

1. Introduction

Polymers are among the most widely utilized materials worldwide, underpinning applications ranging from structural components to functional and responsive systems. However, designing polymers with targeted combinations of properties remains a major challenge. Traditional trial-and-error approaches are time-consuming and costly, while classical statistical methods such as Design of Experiments (DoE) and the Taguchi method become increasingly impractical as the design space grows combinatorially. This challenge is particularly pronounced in monomer and formulation design, where numerous compositional and structural variables must be considered simultaneously. In recent years, machine learning (ML) and data-driven methodologies have emerged as promising tools to accelerate polymer design [1–10]. By learning structure-property relationships from data, these approaches enable efficient screening and optimization beyond what manual intuition can

provide. Nevertheless, their practical deployment in polymer research is severely constrained by data scarcity. Obtaining high-fidelity polymer property data under consistent conditions typically requires extensive experimental cost and time, resulting in datasets that rarely exceed 10^2 samples. In this context, data scarcity is defined as a relative condition in which the combinatorial complexity of the formulation space far exceeds the accessible experimental budget. Although the proposed active-learning framework efficiently navigates this scarcity, its direct transferability to extremely high-dimensional structural spaces remains a subject for further validation, as the required initial sampling density and surrogate-model scalability may become limiting factors. This fundamental limitation has significantly slowed progress in data-driven polymer design relative to other domains.

Among various strategies to address limited data, including few-shot learning, transfer learning, and active learning [1,11–20], active learning is particularly attractive for experimental polymer design because

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it explicitly accounts for the cost of data acquisition and iteratively selects the most informative experiments. Beyond data efficiency, polymer design problems are inherently multi-objective, where trade-offs frequently arise between competing properties such as strength versus ductility or stiffness versus relaxation dynamics. Consequently, the goal of materials optimization is not to identify a single optimal solution but to obtain a set of Pareto-optimal solutions that represent different compromises among objectives [21]. Recent studies have demonstrated the potential of multi-objective optimization (MOO) in polymer systems [22–28]. However, most existing approaches primarily focus on improving optimization performance, while the role of algorithmic design choices in experimental efficiency remains less understood. To address this, recent benchmarking efforts in materials discovery have evaluated the performance of distinct active learning components. For instance, Wang et al. compared various acquisition functions, revealing that their efficacy is highly contingent on the complexity of the materials search space [29]. Similarly, Low et al. benchmarked Bayesian and evolutionary multi-objective strategies, finding that while Bayesian methods such as q-Noise Expected Hypervolume Improvement (qNE-HVI) are sample-efficient for rapid discovery of the Pareto front, they often fail to fully exploit the diversity of optimal solutions compared to evolutionary approaches [30]. Issa et al. further highlighted the limitations of conventional point-wise uncertainty sampling, proposing integrated posterior variance to account for the global impact of candidate selection [31].

However, these studies typically treat individual components of the optimization workflow, such as the sampling method, surrogate model, or acquisition function, in isolation rather than as a coupled system. Therefore, the novelty of this work lies in shifting the focus from optimizing isolated algorithmic components to evaluating their coupled interactions within a unified active learning pipeline. In experimental polymer design, where each data point requires significant time and cost, such simplifications become problematic. Under severe data scarcity, the optimization process cannot rely on iterative self-correction, and suboptimal configurations may lead to premature convergence or inefficient exploration. This observation highlights a critical issue. The efficiency of active learning in experimental polymer design is governed not only by individual algorithmic components but also by the coupling between initialization and acquisition strategies under severe data scarcity. The theoretical justification for this coupling is rooted in how the initial sampling directly shapes the surrogate model's posterior distribution [32]. The initial sampling determines the spatial coverage of the design space and the baseline uncertainty structure of the surrogate model, which directly influences the exploration and exploitation behavior of the acquisition function. Since the acquisition function strictly operates on the probabilistic landscape provided by the initial model, a poorly structured initialization will inevitably mislead the acquisition process, either trapping it in local optima due to over-exploitation or wasting limited budgets on uninformative exploration [31,32]. To reconcile the inherent trade-off between comprehensive design space coverage and severe experimental data scarcity, we allocate a minimal initial budget to Maximin sampling. This pushes the initial formulations toward extreme boundary regions, enabling the surrogate model to operate primarily within an interpolation regime rather than struggling with highly uncertain extrapolation. Consequently, this sparse but broad boundary allows the Gaussian Process model to efficiently direct the acquisition function toward high-performance Pareto regions with minimal subsequent iterations. Despite its importance, this coupling effect remains insufficiently understood in experimentally constrained polymer systems.

To address this gap, we consider two conceptually aligned polymer systems, namely polyurethanes (PU) and polyester-based vitrimers, to evaluate active learning strategies across both algorithmic and experimental settings. The PU dataset, characterized by a highly combinatorial formulation space and heterogeneous property distributions, serves as a literature-derived benchmark for identifying robust combinations of

strategies [33]. Although such literature-derived datasets carry selection biases and heterogeneous property distributions, they serve as rigorous algorithmic stress tests. We do not transfer learned chemical parameters from the PU dataset to the vitrimer system; rather, we transfer the algorithmic strategy itself. That this strategy pairing successfully guides the discovery of optimal formulations in our independent, physically synthesized vitrimer system demonstrates that the framework's efficiency is robust and agnostic to initial literature bias. Building upon these insights, vitrimer systems provide a physically grounded platform for experimental validation [34,35]. Vitrimers are permanently crosslinked networks capable of dynamic bond exchange, combining the mechanical robustness of thermosets with the reprocessability of thermoplastics [36]. In polyester-based vitrimers, formulation variables such as diacid selection and molar ratios strongly influence the trade-off between mechanical stiffness (Young's modulus) and network mobility (stress relaxation time) [37]. This formulation-centric perspective enables a direct assessment of how algorithmic design choices translate into real experimental outcomes.

In this work, we establish an active learning framework for data-efficient multi-objective polymer design. By systematically evaluating combinations of two initialization strategies, namely Maximin and Latin Hypercube Sampling, and three acquisition functions, namely Expected Hypervolume Improvement, exploration and exploitation, and ϵ -Pareto Active Learning, within a Gaussian Process framework, we identify robust strategy pairings for experimental optimization. The results show that substantial Pareto improvements can be achieved within the earliest iterations. Using only four additional experiments, the framework identifies formulations that significantly expand the stiffness-mobility trade-off space. These findings demonstrate that well-designed active learning strategies can accelerate polymer discovery under strict experimental constraints and provide practical guidance for data-scarce materials design.

2. Methods

2.1. Dataset

2.1.1. PU dataset

The PU dataset was obtained from Rapp et al. [33]. In this dataset, a polyurethane (PU) oligomer is composed of up to two diols and two diisocyanates. When two monomers of the same type are used, their molar ratios are restricted to 1:1, 1:3, or 3:1. To control the influence of molecular weight and concentration on network structure and mechanical properties, the molar mass was limited to 2000 g/mol and 4000 g/mol, and two oligomer concentrations (1.5 M and 2.0 M, relative to isocyanate) were considered. The dataset includes six diols and four diisocyanates. The experimentally measured properties are stress and strain at break, yielding 132 data points. Each formulation is encoded as a 12-dimensional vector that represents compositional fractions.

2.1.2. Vitrimer dataset

A polyester-based vitrimer system was selected as the experimental platform to validate the proposed active learning framework. This vitrimer relies on dynamic covalent bond exchange via transesterification reactions and is synthesized by curing an epoxy resin with carboxylic acids or anhydrides in the presence of a catalyst. The epoxy monomer was fixed as diglycidyl ether of bisphenol A (DGEBA). The carboxylic acid components were selected from four aliphatic dicarboxylic acids, namely succinic acid, adipic acid, suberic acid, and sebacic acid. A trifunctional crosslinker, 3-(carboxymethyl)cyclopentane-1,2,4-tricarboxylic acid (TCAA), was introduced to increase crosslink density, and a self-catalytic monomer, di-*n*-butylamine (DBA), was employed as the transesterification catalyst. For each formulation, two diacids were selected and mixed at molar ratios ranging from 0.1 to 0.9. The compositional representation follows the same encoding scheme as the PU dataset to ensure consistency across datasets.

2.2. Surrogate model

Gaussian Process (GP) Regression (GPR) was employed as the uncertainty-aware surrogate model. GPR is a non-parametric Bayesian method that defines a distribution over functions through a mean function and a covariance (kernel) function [38]. This probabilistic formulation enables the prediction of both objective values and associated uncertainties, which are essential for balancing exploration and exploitation in active learning. The GPR models were implemented using the BoTorch library. A Matérn-5/2 kernel with Automatic Relevance Determination (ARD) was adopted. This configuration provides an optimal theoretical compromise for capturing both static properties and dynamic transesterification kinetics, which often exhibit sharp, non-linear transitions that an infinitely differentiable Radial Basis Function (RBF) kernel would over-smooth. ARD further allows the model to learn independent lengthscales for each compositional variable. This kernel choice is empirically supported by a leave-one-out cross-validation (LOOCV) on the initial vitrimer dataset (Supplementary Information Section S3): the Matérn-5/2 kernel attains the lowest predictive RMSE for the dynamic stress relaxation time (2613 s versus 3133 s for RBF), while the corresponding Young's modulus error remains negligible (212 versus 207 MPa). To ensure robust hyperparameter optimization under severe data scarcity, the input composition space was normalized to [0, 1], and the objective outputs were standardized to zero mean and unit variance. This transformation facilitates the reliable use of default hyperparameter priors. The spatial variance maximized by the initial structured sampling strategy effectively regularizes the highly sensitive lengthscale parameters. To account for variations in output magnitude, the kernel was further scaled using a multiplicative scale kernel. Rather than relying on fixed hyperparameters, the GPR model parameters, including the ARD lengthscales, output scale, and observation noise variance, were dynamically optimized at every active learning iteration. This was achieved by maximizing the Exact Marginal Log-Likelihood (MLL) using the `fit_gpytorch_mll` utility in the BoTorch library, ensuring the surrogate model continuously adapted to the newly acquired experimental data.

2.3. Initial dataset sampling

The selection of initial training data plays a critical role in the performance of Bayesian optimization. Prior work has shown that structured initialization strategies can significantly influence optimization trajectories compared to random sampling [32]. To evaluate this effect in multi-objective polymer design, two representative space-filling strategies were considered, namely maximin sampling and Latin Hypercube Sampling (LHS). Based on a comprehensive sensitivity analysis evaluating varying sample sizes (detailed in Supplementary Information Fig. S1), the initial dataset size was fixed at 12 points. Specifically, we evaluated initial sizes of 8, 12, 16, and 20 points across 14 independent random seeds; 12 points yielded a high final hypervolume with the lowest variance (SEM), whereas larger sizes incurred substantially higher upfront cost for diminishing or even slightly negative returns.

- **Maximin Sampling:** This distance-based strategy maximizes the minimum Euclidean distance between all sampled points, promoting coverage of boundary and extreme regions in the design space.
- **Latin Hypercube Sampling (LHS):** LHS ensures uniform marginal distributions across input variables. For the discrete formulation space considered here, continuous LHS samples were mapped to the nearest realizable compositions.

2.4. Active learning

The active learning workflow is illustrated in Fig. 1. The framework is designed to navigate high-dimensional design spaces with high sample efficiency by iteratively updating a surrogate model and querying

informative candidates. At each iteration, the Gaussian Process model is refined using cumulative experimental data, providing a probabilistic representation of the objective space that captures both predicted performance and epistemic uncertainty. The workflow consists of four stages:

1. **Design Space Construction:** This stage defines the chemical components, compositional constraints, and target material properties, thereby establishing the combinatorial design space and optimization objectives.
2. **Structured Initialization:** A space-filling sampling strategy is employed to generate an initial training set. This step ensures sufficient coverage of the design space, reducing extrapolation and enabling the surrogate model to operate primarily within an interpolation regime.
3. **Surrogate Modeling and Candidate Querying:** The accumulated dataset is used to train a Gaussian Process surrogate model. The acquisition function is then applied to the candidate pool to evaluate both expected performance and uncertainty, thereby selecting formulations that maximize information gain or Pareto improvement.
4. **Experimental Validation and Iteration:** The selected candidates are synthesized and characterized experimentally. The newly obtained data are incorporated into the training set, and the surrogate model is updated for the next iteration. The process continues until the stopping criterion is met, namely when no candidate in the newly queried batch increases the dominated hypervolume. Since termination requires an entire batch to fail simultaneously, the criterion is mildly sensitive to the batch size: larger batches tend to postpone termination, whereas smaller batches stop earlier. A batch size of 4 was adopted to balance per-iteration experimental throughput against the risk of premature stopping.

2.4.1. Multi-objective acquisition strategies

In multi-objective optimization (MOO), the objective shifts from identifying a single optimum to reconstructing the Pareto front. While numerous MOO algorithms have been proposed, evaluating minor mathematical variants often yields limited practical insight for experimental applications. To provide a clear and representative comparison, we consider three acquisition strategies corresponding to distinct decision-making paradigms in materials discovery, namely metric-based quantification, classification-oriented partitioning, and heuristic-based alternation.

- **Expected Hypervolume Improvement (EHVI):** EHVI represents the metric-based paradigm and is widely regarded as a standard approach in probabilistic multi-objective optimization. It selects the candidate expected to provide the largest improvement in dominated hypervolume relative to the current Pareto set. Let $D_t = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{n_t}$ denote the accumulated dataset at iteration t , and let the Gaussian Process surrogate provide a predictive distribution for a candidate formulation $\mathbf{x}$:

$$\mathbf{f}(\mathbf{x}) \mid D_t \sim \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}), \boldsymbol{\Sigma}_t(\mathbf{x})), \quad (1)$$

where $\boldsymbol{\mu}_t(\mathbf{x})$ and $\boldsymbol{\Sigma}_t(\mathbf{x})$ are the posterior mean vector and covariance matrix of the objectives, respectively. Given the current non-dominated objective set P_t and a reference point $\mathbf{r}$, the hypervolume improvement produced by a sampled objective vector $\mathbf{y}$ - where $HV(\cdot, \mathbf{r})$ denotes the hypervolume function defined formally in Section 2.5 - is defined as:

$$I_{HV}(\mathbf{y}; P_t, \mathbf{r}) = HV(P_t \cup \{\mathbf{y}\}, \mathbf{r}) - HV(P_t, \mathbf{r}). \quad (2)$$

The EHVI acquisition function is then given by:

$$\alpha_{EHVI}(\mathbf{x}) = \mathbb{E}_{\mathbf{f}(\mathbf{x}) \mid D_t} [I_{HV}(\mathbf{f}(\mathbf{x}); P_t, \mathbf{r})]. \quad (3)$$

The next candidate is selected by maximizing EHVI over the candidate pool $\mathcal{X}_{\text{cand}}$:

$$\mathbf{x}_{t+1} = \arg \max_{\mathbf{x} \in \mathcal{X}_{\text{cand}}} \alpha_{EHVI}(\mathbf{x}). \quad (4)$$

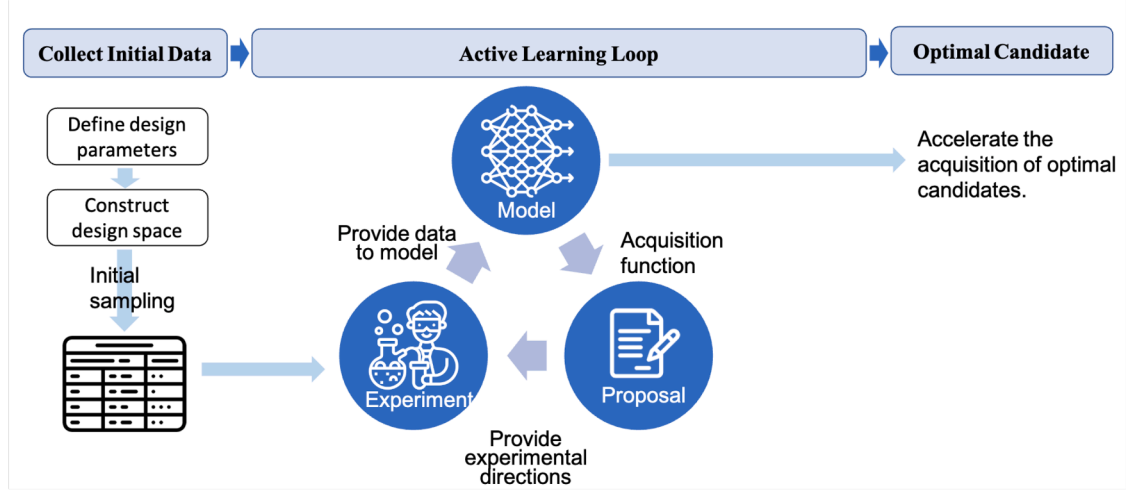


Fig. 1. Overview of the active learning framework. Computational stages, including design space construction, surrogate modeling, and acquisition optimization, are shown in green, while experimental synthesis and validation steps are shown in red. The iterative loop progressively updates the dataset by selecting informative formulations, enabling efficient exploration of the design space and reconstruction of the Pareto frontier. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

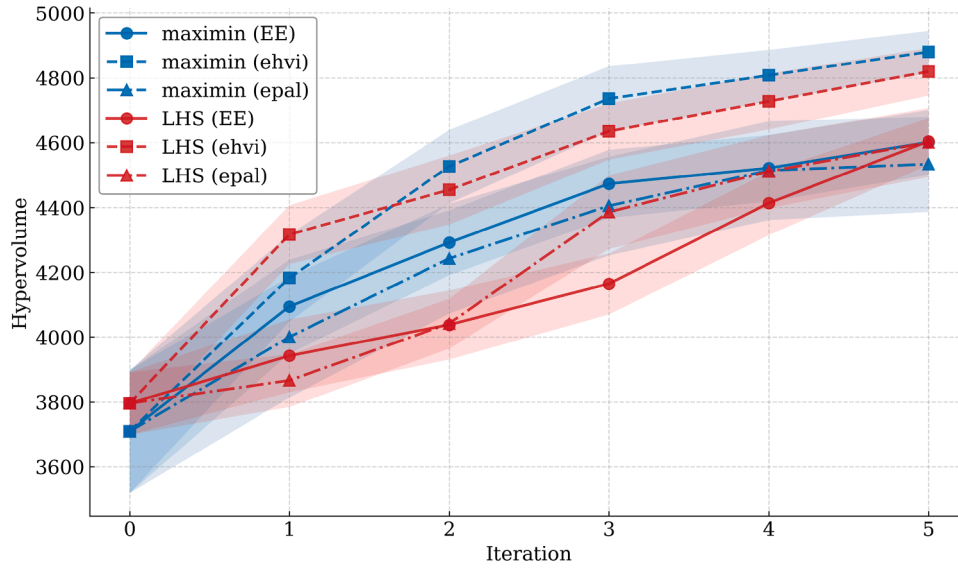


Fig. 2. Evolution of hypervolume across successive active learning iterations on the PU dataset. Solid lines denote E&E, dashed lines denote EHVI, and dash-dot lines denote ϵ -PAL acquisition strategies. Blue curves represent Maximin initialization, while red curves indicate Latin Hypercube Sampling (LHS). Shaded regions denote the standard error of the mean (SEM) calculated from 14 independent trials. A performance crossover is observed, where Maximin + EHVI rapidly overtakes the initially higher LHS starting point. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This formulation naturally balances convergence and diversity because candidates are rewarded only when they are expected to expand the dominated hypervolume. In practice, the EHVI integral is evaluated using the Monte Carlo q -Noisy Expected Hypervolume Improvement (qNEHVI) estimator implemented in BoTorch, which provides a tractable, noise-robust approximation.

- **ϵ -Pareto Active Learning (ϵ -PAL):** ϵ -PAL represents the classification-oriented paradigm by formulating MOO as a partitioning problem [23]. It iteratively classifies the design space into Pareto-optimal, non-Pareto, and uncertain regions based on a user-defined tolerance parameter ϵ . This approach enables robust identification of Pareto candidates under uncertainty.
- **Exploration and Exploitation (E&E):** E&E represents a heuristic-based paradigm commonly used in experimental practice. It alternates between uncertainty-driven exploration and Pareto-based exploitation. For a general bi-objective optimization problem, let $\mathbf{x}$ de-

note a candidate formulation in the candidate pool $\mathcal{X}_{\text{cand}}$, and let the two target properties be represented as $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$. At iteration t , the Gaussian Process surrogate provides posterior mean predictions:

$$\boldsymbol{\mu}_t(\mathbf{x}) = [\mu_{t,1}(\mathbf{x}), \mu_{t,2}(\mathbf{x})], \quad (5)$$

where $\mu_{t,1}(\mathbf{x})$ and $\mu_{t,2}(\mathbf{x})$ denote the predicted mean values of Property 1 and Property 2, respectively. The corresponding posterior standard deviation vector is defined as:

$$\boldsymbol{\sigma}_t(\mathbf{x}) = [\sigma_{t,1}(\mathbf{x}), \sigma_{t,2}(\mathbf{x})], \quad (6)$$

where $\sigma_{t,1}(\mathbf{x})$ and $\sigma_{t,2}(\mathbf{x})$ quantify the predictive uncertainties associated with property 1 and property 2, respectively. During the exploration step, candidates with the largest predictive uncertainty are selected. The overall predictive uncertainty of a candidate is quantified by the Euclidean norm of the posterior standard deviation vector:

$$U_t(\mathbf{x}) = \|\boldsymbol{\sigma}_t(\mathbf{x})\|_2 = \left(\sigma_{t,1}^2(\mathbf{x}) + \sigma_{t,2}^2(\mathbf{x}) \right)^{1/2}. \quad (7)$$

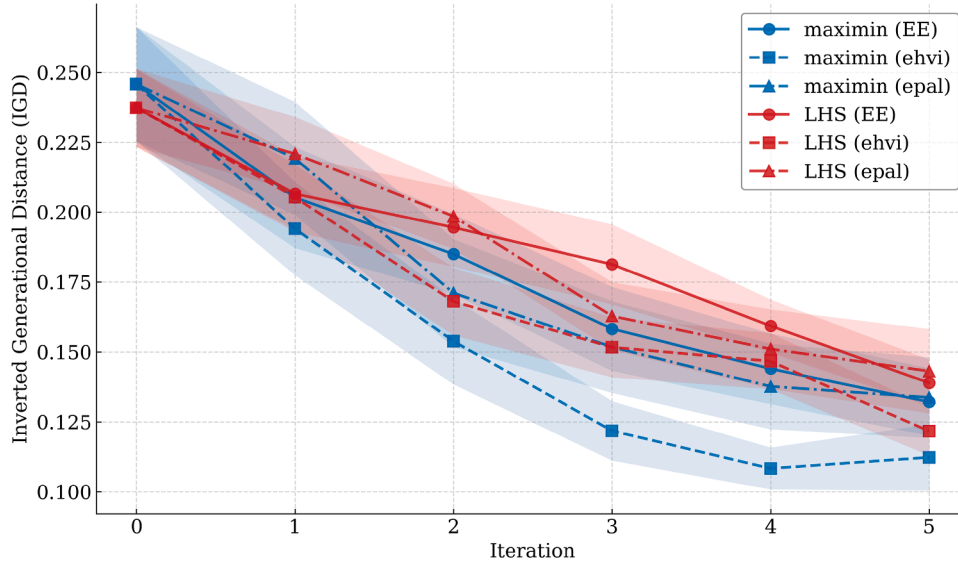


Fig. 3. Evolution of the Inverted Generational Distance (IGD) across active learning iterations for the PU dataset. Lower values indicate a closer, more uniformly distributed approximation to the true Pareto front. Shaded regions represent the standard error of the mean (SEM) derived from 14 independent random seeds. The Maximin + EHVI strategy achieves the most rapid and stable convergence.

The exploration batch is then selected as the top- K candidates with the largest uncertainty values:

$$B_t^{\text{explore}} = \text{TopK}_{\mathbf{x} \in \mathcal{X}_{\text{cand}}} U_t(\mathbf{x}), \quad K = 4. \quad (8)$$

Here, B_t^{explore} denotes the batch selected during the exploration step, and K is the batch size. During the exploitation step, the posterior mean predictions are used to identify the predicted Pareto set:

$$\hat{P}_t = \text{ND}(\{\mu_t(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}_{\text{cand}}\}), \quad (9)$$

where $\text{ND}(\cdot)$ denotes the non-dominated subset under the maximization of both objectives, and $\hat{P}_t$ represents the predicted Pareto set at iteration t . The exploitation batch is then selected from the predicted Pareto set according to a property-prioritized ranking. In the general case, if property 1 is assigned priority, the candidates in $\hat{P}_t$ are ranked according to $\mu_{t,1}(\mathbf{x})$, and the top- K candidates are selected:

$$B_t^{\text{exploit}} = \text{TopK}_{\mathbf{x} \in \mathcal{X}_{\text{cand}}} \{\mu_{t,1}(\mathbf{x}) \mid \mu_t(\mathbf{x}) \in \hat{P}_t\}, \quad K = 4. \quad (10)$$

Here, B_t^{exploit} denotes the batch selected during the exploitation step. If property 2 is instead assigned priority, $\mu_{t,1}(\mathbf{x})$ can be replaced by $\mu_{t,2}(\mathbf{x})$ in the ranking criterion. The final E&E selection rule alternates between exploration and exploitation:

$$B_t = \begin{cases} B_t^{\text{explore}}, & t \equiv 1 \pmod{2}, \\ B_t^{\text{exploit}}, & t \equiv 0 \pmod{2}, \end{cases} \quad (11)$$

where B_t is the final selected batch at iteration t . The notation $t \equiv 1 \pmod{2}$ denotes odd-numbered iterations, whereas $t \equiv 0 \pmod{2}$ denotes even-numbered iterations. Thus, the E&E strategy alternates between uncertainty-driven exploration and Pareto-based exploitation throughout the active learning process.

The implementations of EHVI and ϵ -PAL follow the methodologies described in [27] and [23], respectively. For the E&E approach, a simple alternating heuristic is adopted. By comparing these fundamentally different acquisition strategies, we aim to evaluate which approach is most effective under the noise and data scarcity conditions typical of laboratory-generated polymer datasets.

2.5. Evaluation metric

To comprehensively quantify optimization performance, evaluating both convergence and diversity, two metrics were employed: Hypervolume (HV) [39] and Inverted Generational Distance (IGD). Hypervolume

measures the Lebesgue measure (Λ) of the objective space region dominated by the current Pareto set P , bounded by a reference point $\mathbf{r}$. It is defined as:

$$HV(P, \mathbf{r}) = \Lambda\left(\bigcup_{\mathbf{x} \in P} [\mathbf{r}, \mathbf{f}(\mathbf{x})]\right) \quad (12)$$

This metric simultaneously reflects convergence toward the true Pareto front and the diversity of the trade-off solutions. In this work, HV is computed after each iteration to track the evolution of the sampled Pareto front. To ensure consistency across iterations and initialization strategies, the reference point must be strictly worse than all attainable objective values. For the PU dataset benchmark, the reference point was fixed at $\mathbf{r} = (0, 0)$. While HV is robust, it can be sensitive to the choice of the reference point and may sometimes obscure the specific balance between convergence and diversity. To rigorously address this limitation, the Inverted Generational Distance (IGD) was introduced as a secondary evaluation metric. IGD calculates the average Euclidean distance from every point on the true Pareto front (P^*) to its nearest neighbor in the algorithmically obtained Pareto front (P):

$$IGD(P, P^*) = \frac{1}{|P^*|} \sum_{\mathbf{v} \in P^*} \min_{\mathbf{u} \in P} d(\mathbf{v}, \mathbf{u}) \quad (13)$$

where $d(\mathbf{v}, \mathbf{u})$ represents the Euclidean distance in the objective space. To prevent objectives with larger numerical ranges from dominating the distance calculation, all objective values were normalized to a $[0, 1]$ scale prior to calculating $d(\mathbf{v}, \mathbf{u})$. A lower IGD value indicates superior performance, confirming that the obtained solutions are both highly accurate and well-distributed.

2.6. Vitrimers experiment detail

2.6.1. Materials

Diglycidyl ether of bisphenol A (DGEBA, epoxy equivalent weight ca. 170 g/eq) was purchased from Sigma-Aldrich (St. Louis, MO, USA). Succinic acid (SA) was obtained from ECHO Chemical Co., Ltd. (Miaoqi, Taiwan). Adipic acid (AA) and sebacic acid (SeA) were procured from Anhui Sunsing Chemicals Co., Ltd. (Anhui, China). Suberic acid (SuA) and di-*n*-butylamine (DBA) were purchased from UNI-ONWARD Corp. (New Taipei City, Taiwan). The trifunctional crosslinker, 3-(carboxymethyl)cyclopentane-1,2,4-tricarboxylic acid (TCAA), was

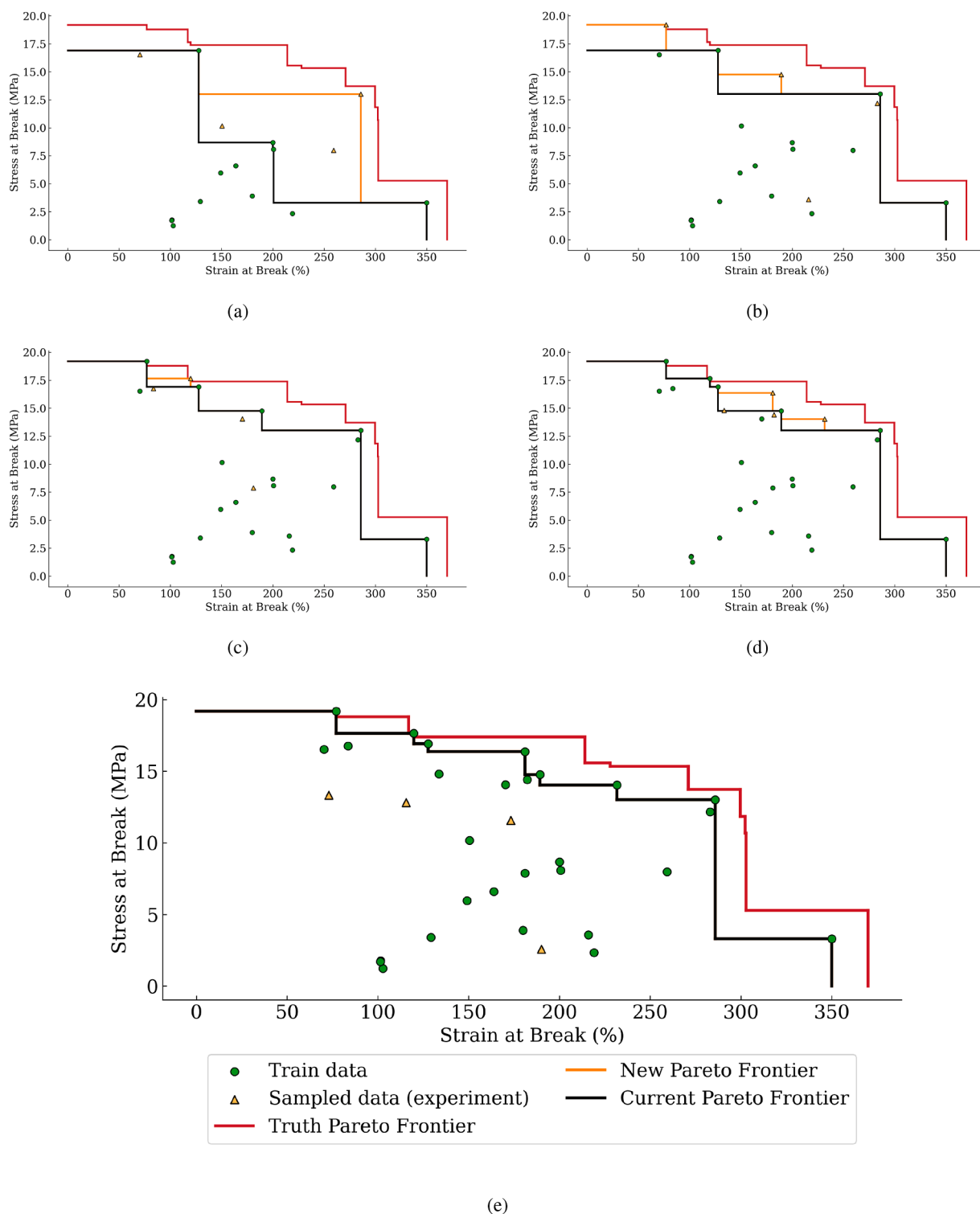


Fig. 4. Evolution of the sampled Pareto front across five Maximin + EHVI iterations on the PU dataset. The red line denotes the ground truth Pareto front. Green dots represent cumulative sampled formulations. Black and yellow lines indicate the Pareto front before and after each iteration, respectively, and yellow triangles denote newly selected candidates. The results illustrate progressive expansion of the Pareto front and convergence by the fifth iteration. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

obtained from LCY Chemical Corporation (Taipei City, Taiwan). All chemicals were used as received without further purification.

2.6.2. Synthesis of vitrimer

Polyester-based vitrimer samples were prepared via a one-pot melt mixing process. DGEBA was used as the epoxy monomer. A series of

aliphatic dicarboxylic acids with varying methylene chain lengths, including succinic acid (C_4), adipic acid (C_6), suberic acid (C_8), and sebacic acid (C_{10}), were employed as polyester-forming diacids. TCAA was used as a trifunctional crosslinker to increase network crosslink density, while DBA served as the catalyst for transesterification. In all formulations, the molar ratio of carboxyl groups to epoxy groups was fixed

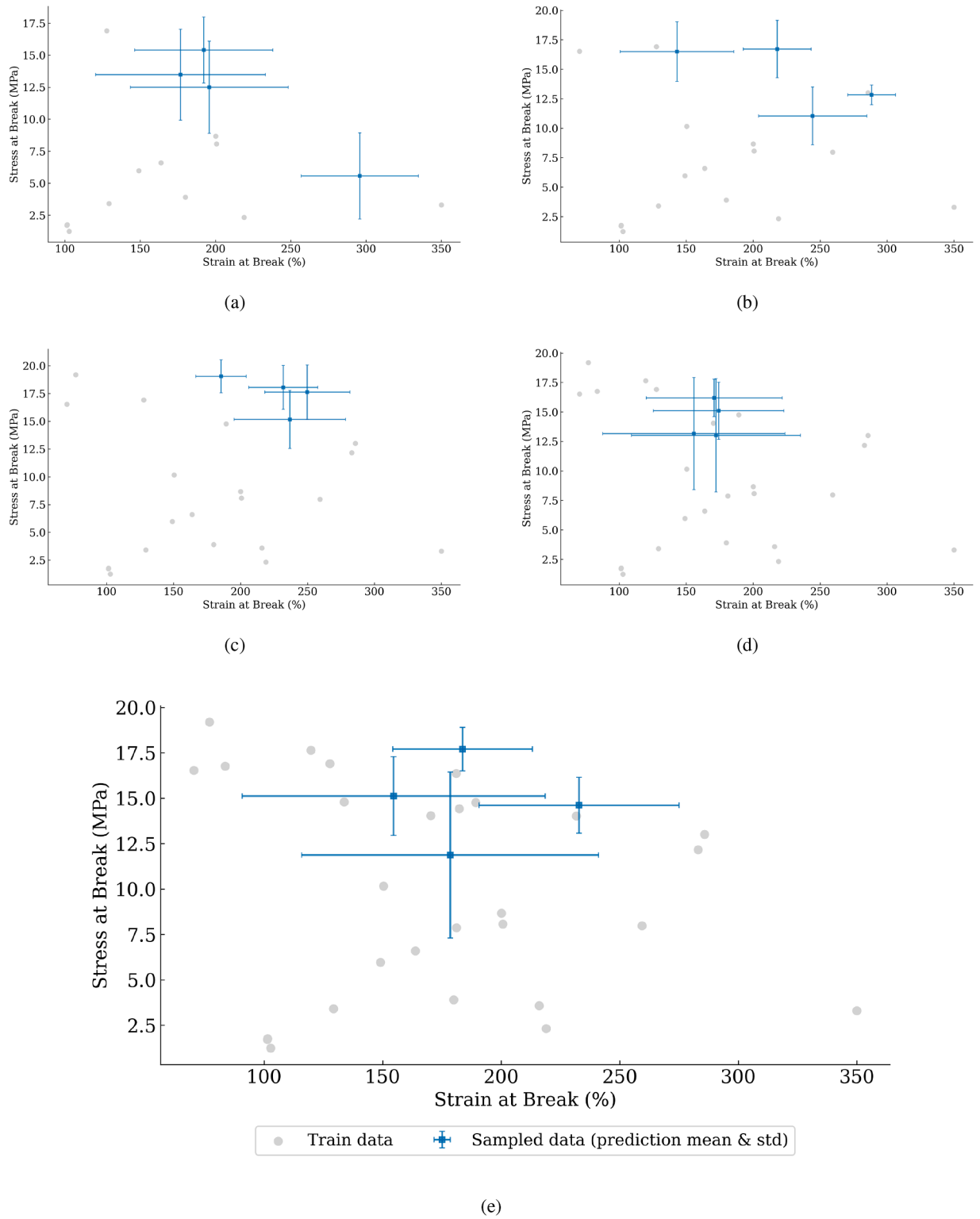


Fig. 5. Evolution of predicted candidates and their associated uncertainties across five active learning iterations for the PU dataset. Light-gray points represent previously sampled data. Blue points denote newly selected candidates, and error bars indicate predictive standard deviations from the surrogate model. The results illustrate how candidate selection evolves from uncertainty-driven exploration to performance-driven refinement as the model converges. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

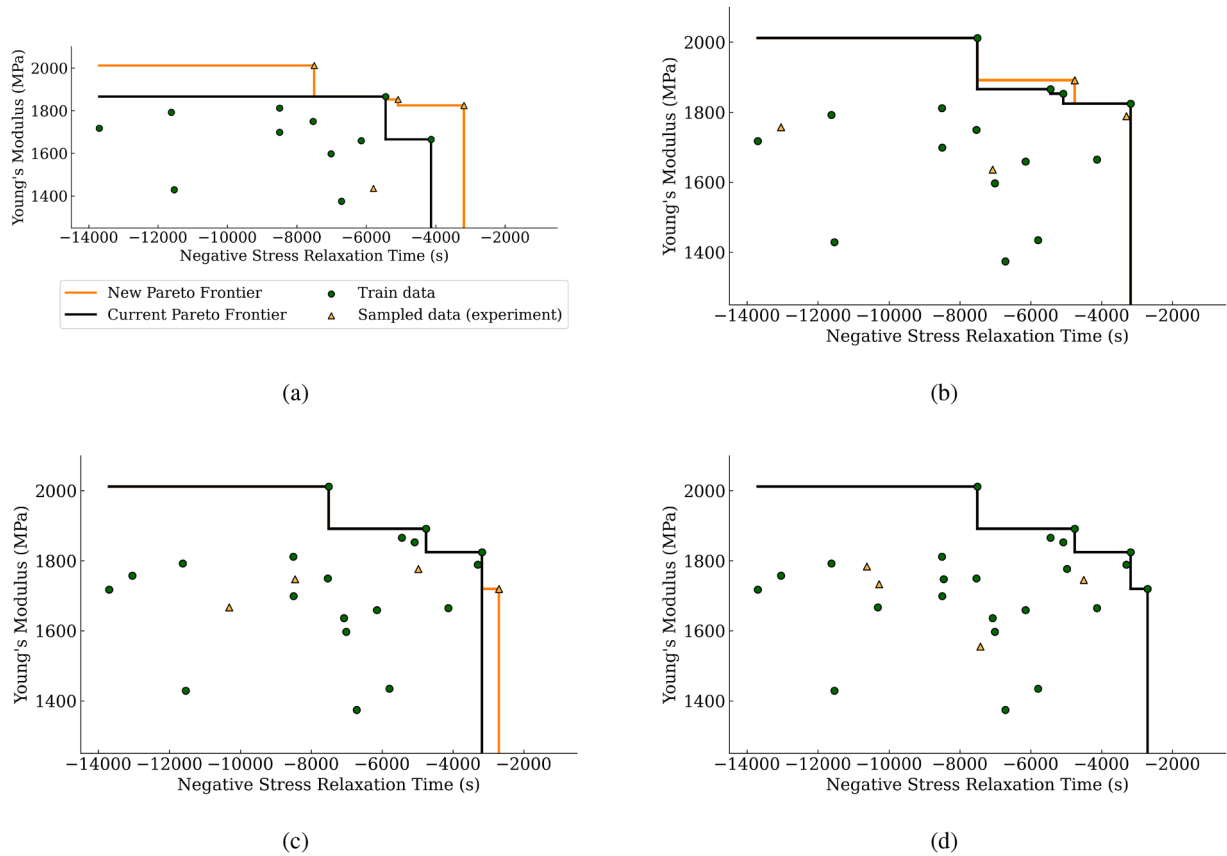


Fig. 6. Evolution of the experimental Pareto front across four active learning iterations for the polyester-based vitrimer system. Green dots represent cumulative sampled formulations. Black lines denote the Pareto front before each iteration, while yellow lines indicate the updated front after incorporating newly selected candidates (yellow triangles). The results show initial expansion of the Pareto front, followed by refinement and convergence as the optimization progresses. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

at 1:1. In a typical synthesis, DGEBA, two selected diacids, and TCAA were mixed in a polytetrafluoroethylene (PTFE) beaker and heated to 160 °C under stirring until a homogeneous melt was obtained. DBA was then added and uniformly dispersed. Curing was conducted in two steps. Samples were pre-cured at 110 °C for 1 h to induce gelation, followed by post-curing at 160 °C for 6 h to complete network formation. The samples were cooled to room temperature and demolded for subsequent testing.

2.6.3. Characterizations

Two key properties were characterized, namely Young's modulus and stress relaxation time. Uniaxial tensile tests were performed to determine Young's modulus. Dog-bone specimens were prepared according to ASTM D638 Type V and tested at a crosshead speed of 10 mm/min using a universal testing machine (Cometech QC-508M2F, Taichung City, Taiwan) at approximately 28 °C with a gauge length of 9.53 mm. Young's modulus was obtained from a linear regression of the stress-strain curve over the 5th-15th data points. To ensure reproducibility and explicitly quantify measurement uncertainty, the tensile tests for each validated formulation (both Pareto- and non-Pareto-front) were conducted on 5 independent replicate specimens. The testing instrumentation was systematically calibrated prior to measurements to minimize systematic instrumental errors. Stress relaxation behavior was measured using a rotational rheometer (Anton Paar MCR 302) in torsion mode under isothermal conditions. Rectangular specimens with dimensions 2 × 5 mm² and a clamp distance of 20 mm were subjected to a constant angular deformation of 30°. The stress relaxation time τ was defined as the time required for the torque to decay to 0.37 of its initial value. To optimize experimental efficiency and minimize characterization time, 3

repeated rheological measurements were conducted on formulations on the Pareto front, while non-Pareto-front formulations were characterized with a single stress-relaxation measurement due to the long duration of each rheological run. The instrument's thermal and torque calibrations were verified prior to each batch. The resulting property values for the Pareto-front samples are reported as the mean $\pm$ standard deviation (SD).

3. Results

3.1. Comparison of different strategies

To rigorously quantify the efficacy of different active learning configurations and eliminate the bias of random initialization, we monitored the evolution of HV and IGD across 14 independent trials with varying random seeds. The averaged evolutionary trajectories, along with the standard error of the mean (SEM), are presented in Figs. 2 and 3. A clear performance crossover is observed between the two structured initialization strategies. In the initial state, LHS yields a higher average hypervolume and a slightly lower IGD than Maximin initialization. This initial advantage arises because LHS primarily samples the interior of the compositional space, generating intermediate formulations that tend to exhibit moderate, well-balanced properties. In contrast, the Maximin criterion explicitly maximizes spatial distances, forcing initial observations toward the physical boundaries and extrema of the formulation space. These extreme formulations frequently exhibit one-sided physical properties, resulting in a lower initial hypervolume. However, as the active learning loop progresses, a combination of Maximin initialization and the EHVI acquisition function rapidly overtakes all other configu-

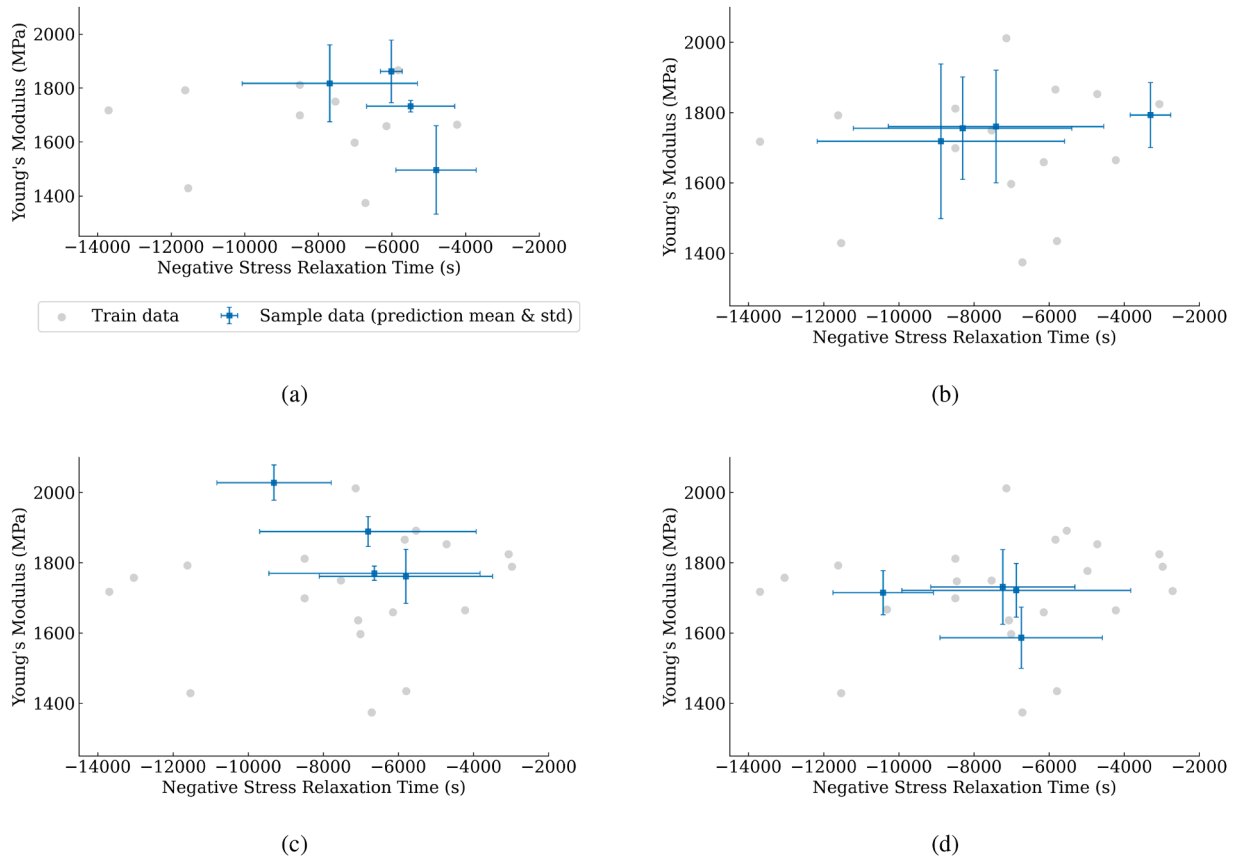


Fig. 7. Evolution of predicted candidates and their associated uncertainties across four active learning iterations for the vitrimer system. Light-gray points represent previously sampled data, while blue points denote newly selected candidates. Error bars indicate predictive standard deviations from the surrogate model. The results show how candidate selection evolves from uncertainty-driven exploration to refinement near the Pareto front as the model converges. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rations. By empirically establishing the absolute limits of the structure-property landscape from the outset, Maximin effectively circumvents the highly uncertain extrapolation that plagues the Gaussian Process surrogate when boundary data is absent. Consequently, EHVI can reliably interpolate within the design space, accurately allocating queries to regions with the highest expected contribution. This synergistic coupling leads to an accelerated convergence trajectory. As demonstrated in the IGD evolution (Fig. 3), which heavily penalizes algorithms that fail to discover the true boundary limits, Maximin + EHVI exhibits a sharp decline in error immediately after the first iteration. While LHS + EHVI eventually catches up as both strategies converge near the true Pareto front at Iteration 5, Maximin + EHVI demonstrates a statistically significant superiority during the critical intermediate learning phase. Reaching the optimal Pareto front 1 or 2 iterations earlier translates to substantial savings in experimental cost, highlighting the sample efficiency of this framework under severe data scarcity (detailed statistical validation in Supplementary Information Section S2). The evaluation also reveals that the EHVI acquisition strategy consistently outperforms both the classification-oriented ϵ -PAL and the heuristic Exploration and Exploitation (E&E) approaches, regardless of the initialization method. The E&E strategy consistently underperforms because its reliance on manual switching between exploration and exploitation introduces instability. Variance-based exploration often selects extreme regions that may lack physical relevance, while exploitation based on an inaccurate early-stage surrogate model reinforces local optima driven by noise. These results confirm that acquisition strategies such as EHVI, which probabilistically quantify the exact contribution of each candidate to the expected expansion of the Pareto front, are structurally more effective for

navigating data-scarce experimental polymer design. Under an identical total budget of 32 samples, the Maximin + EHVI strategy attains a substantially higher final hypervolume (4880.25 ± 64.82) than a naive Random Search baseline (4702.62 ± 65.57), confirming that the observed gains stem from informed candidate selection rather than sampling volume alone (Supplementary Information Section S1).

3.2. Quantitative analysis of Pareto expansion

To evaluate the efficiency of the Maximin + EHVI approach, we tracked the incremental expansion of the Pareto front over five iterations. In the first iteration, a newly selected formulation expanded the central region of the Pareto front, as shown in Fig. 4a. The initial dataset exhibited a pronounced trade-off between strength and ductility: the formulation with the highest tensile stress (16.91 MPa) showed a relatively low strain of 127.7%, whereas the most ductile formulation reached 350% strain with a low stress of 3.3 MPa. A single EHVI-guided experiment identified a formulation with a stress of 5.56 MPa and a strain of 295.83%. This corresponds to a 68% increase in stress at a comparable strain level, effectively narrowing the gap between high-strength and high-ductility regimes. As the iterations progressed (Fig. 4b–d), the Pareto front continued to move toward higher-performance regions. By the fifth iteration, newly sampled points no longer extended the Pareto front, indicating that the accessible design space had been sufficiently explored and that the stopping criterion was satisfied. These results highlight the ability of the Maximin + EHVI strategy to achieve substantial Pareto improvements within very limited experimental budgets.

3.3. Rationalization of candidate selection

The selection of candidates can be interpreted in terms of the probabilistic landscape defined by the Gaussian Process Regression surrogate and the EHVI acquisition logic. The selected formulations can be categorized into three groups based on their predictive mean and uncertainty:

- **High-uncertainty group:** Formulations located in sparsely sampled regions with high predictive variance. These candidates drive **exploration** by reducing model uncertainty and potentially revealing new Pareto-optimal regions.
- **High-predicted-value group:** Formulations predicted to lie on or beyond the current Pareto front with relatively low uncertainty. These correspond to **exploitation**, refining the local structure of the Pareto front.
- **Synergistic group:** Formulations that exhibit both high predicted performance and high uncertainty. These candidates are expected to contribute most effectively to expanding the Pareto front.

To further analyze the selection behavior, the predicted means of the candidate formulations were visualized in the objective space, with error bars indicating the predictive standard deviations. The evolution of candidate selection across iterations reflects the adaptive behavior of the framework. In the first iteration (Fig. 5a), selection is dominated by synergistic candidates, reflecting the high uncertainty of the initial surrogate model. With limited training data, predicted performance and uncertainty are strongly coupled, leading to selections that simultaneously explore and improve the Pareto front. As the dataset expands (Fig. 5b and c), the overall predictive uncertainty decreases. During this stage, the acquisition strategy shifts toward exploitation, prioritizing candidates with higher predicted performance to refine the Pareto front. By the fourth iteration (Fig. 5d), a mixed selection pattern emerges. Some candidates are chosen to extend the Pareto front, while others probe under-sampled regions. This behavior demonstrates that EHVI adaptively balances exploration and exploitation in response to an evolving landscape of uncertainty. The combination of Maximin initialization and EHVI acquisition provides a consistent advantage for multi-objective polymer design under severe data scarcity. Maximin ensures boundary coverage of the design space, while EHVI allocates queries according to their expected contribution to Pareto front expansion, resulting in high sample efficiency.

3.4. Experimental demonstration on vitrimer formulations

To evaluate the effectiveness and generality of the proposed active learning framework, we applied it to a polyester-based vitrimer system. Whereas the PU benchmark in Section 3 enabled statistical evaluation across 14 independent random seeds, the vitrimer demonstration reported here corresponds to a single physical active-learning campaign; the improvements reported below therefore reflect the realized trajectory of one such campaign rather than an average across repeated runs.

3.4.1. Quantitative analysis

Fig. 6 illustrates the evolution of the Pareto front across four active learning iterations. In the first iteration, the framework expanded the experimental Pareto front by targeting boundary regions of the design space. Compared with the initial 12 formulations generated by Maximin sampling, the four selected candidates resulted in improvements in both the high-stiffness and high-mobility regimes. In the high-modulus region, Young's modulus increased from 1865 ± 101 MPa to 2011 ± 125 MPa (a 7.8% gain). In the fast-relaxation regime, the stress relaxation time decreased from 4131 ± 151 s to 2706 ± 33 s (a 34.5% reduction), while Young's modulus simultaneously increased from 1664 ± 55 MPa to 1719 ± 44 MPa (a 3.3% gain).

These results indicate a substantial expansion of the Pareto front within a single iteration. The ability to identify formulations that improve both objectives highlights the efficiency of the Maximin + EHVI

Table 1

Comparison of formulation compositions and measured properties for the Pareto-optimal vitrimers identified in the initial dataset (Iteration 0) and the final active learning stage (Iteration 4). Values are reported as Mean $\pm$ SD.

Stage	Composition	Young's Modulus (MPa)	Relaxation Time (s)
Initial Pareto Front	SA:AA = 4:6	1865 ± 101	5441 ± 339
	SA:SeA = 4:6	1664 ± 55	4131 ± 151
Final Pareto Front	SA:AA = 5:5	1719 ± 44	2706 ± 33
	SA:SeA = 5:5	2011 ± 125	7502 ± 624
	AA:SuA = 2:8	1891 ± 58	4758 ± 674
	AA:SeA = 2:8	1824 ± 37	3185 ± 1474

strategy under limited experimental budgets. In the second iteration, the rate of Pareto expansion decreased, and the framework shifted toward refining the central trade-off region. A candidate identified at this stage (Fig. 6b) exhibited an improved balance between stiffness and network mobility that was absent from the initial dataset. By the third and fourth iterations (Fig. 6c–d), hypervolume growth showed diminishing returns. In the third iteration, only one candidate slightly extended the Pareto front, whereas in the fourth iteration, no candidates improved it. This behavior indicates that the accessible design space had been sufficiently explored and that the stopping criterion had been met. The final Pareto set provides a broader and more continuous range of property combinations than the initial dataset, offering practical design options from compliant, fast-relaxing networks to stiff, slow-relaxing thermosets. A notable feature of this optimization trajectory is the large improvement achieved in the first iteration. This contrasts with the commonly observed cold-start problem in active learning, where early iterations provide limited benefit [40–42]. In typical Bayesian optimization workflows, several iterations are required to reduce model uncertainty before meaningful improvements are observed. In contrast, Maximin initialization ensures coverage of boundary regions from the outset, reducing extrapolation and enabling the surrogate model to operate within a more reliable interpolation regime. When combined with EHVI, which prioritizes candidates based on their expected contribution to the expansion of the Pareto front, the framework rapidly identifies high-impact formulations. With only four additional experiments, a 34.5% improvement in relaxation time was achieved in this experimental campaign, demonstrating high sample efficiency under experimental constraints.

To explicitly demonstrate the compositional shifts and property enhancements achieved through the active learning process, Table 1 summarizes the chemical formulations and corresponding measured properties of the Pareto-optimal candidates identified in the initial Maximin sampling stage versus those forming the final expanded Pareto front.

3.4.2. Model interpretation

Consistent with the candidate groupings established in the PU dataset, the predicted means and standard deviations of the selected vitrimer candidates exhibit a similar pattern. Their distributions in the objective space are shown in Fig. 7. In the first iteration (Fig. 7a), the selected candidates can be divided into two categories: a high-predicted-value group and a high-uncertainty group. The light-gray points represent previously sampled data, while blue points denote newly selected candidates. Two candidates are located near the current Pareto front and have relatively low uncertainty, thereby refining the existing boundary. The remaining candidates are positioned in dominated regions with higher uncertainty, indicating exploration of under-sampled areas. These exploration-driven selections correspond to the observed reduction in stress relaxation time reported in the quantitative analysis, suggesting that uncertainty-aware sampling helps identify improved formulations. During the second and third iterations (Fig. 7b and c), predictive uncertainty decreases as more data are incorporated. Candidate selection shifts toward exploitation, with more samples located near the Pareto front to refine the trade-off boundary. By the fourth iteration (Fig. 7d), the selection pattern becomes more balanced. Some

candidates are selected near the Pareto front, while others probe regions with relatively higher uncertainty. This behavior indicates that EHVI adaptively balances exploration and exploitation in response to an evolving landscape of uncertainty. This transition from boundary exploration to interior refinement explains the effectiveness of the Maximin + EHVI combination under limited data conditions. Maximin initialization provides coverage of boundary regions, reducing extrapolation in the surrogate model. EHVI then allocates queries according to their expected contribution to Pareto front expansion. Together, these components enable efficient exploration of the design space with a limited experimental budget. In experimental polymer design, where datasets typically contain fewer than 10^2 samples due to synthesis and characterization costs [43,44], such efficiency is critical. Conventional design strategies may allocate resources to regions with limited impact, while purely exploitative approaches risk convergence to local optima [45–48]. In contrast, the proposed framework provides a balanced and data-efficient approach for identifying high-performance formulations.

4. Discussion

The results demonstrate that coupling structured Maximin initialization with the EHVI acquisition function efficiently accelerates multi-objective polymer design under severe data scarcity. However, to establish a balanced perspective on the practical deployment of this data-driven framework, it is crucial to explicitly acknowledge its inherent limitations and define the boundaries of its applicability.

4.1. Theoretical rationalization of strategy coupling

The empirical success of coupling Maximin initialization with the EHVI acquisition function, as observed in both the PU benchmark and the experimental vitrimer system, can be fundamentally rationalized through the mathematical interplay between spatial geometry, predictive uncertainty, and acquisition logic. Firstly, from the perspective of experimental design geometry, Maximin mathematically guarantees superior boundary coverage compared to LHS [49]. While LHS ensures one-dimensional marginal uniformity, it frequently leaves multidimensional geometric voids, particularly at the boundaries of the design space. By contrast, the Maximin criterion explicitly maximizes the minimal Euclidean distance between sampled points, systematically forcing initial observations toward the physical extrema of the formulation space. Secondly, this geometric boundary coverage directly governs the predictive reliability of the Gaussian Process surrogate. According to GP theory [50,51], predictive variance increases critically in extrapolation regimes, areas outside the convex hull of the training data. By anchoring the extremes of the structural-property landscape early in the process, Maximin initialization structurally transforms highly uncertain extrapolation regions into interpolation regions. Finally, the exact computation of EHVI relies strictly on the accuracy of this predictive variance. EHVI integrates over the GP posterior to estimate expected hypervolume gains [52]. If the boundary variance is artificially inflated, the acquisition function will severely overestimate the expected gains in those unexplored boundaries. By mathematically bounding the predictive variance via Maximin sampling, EHVI is theoretically constrained to query regions with genuine Pareto-expansion potential. Consequently, this coupling effectively prevents wasted experimental budget and accelerates convergence, a mechanism that is indispensable under severe data scarcity.

4.2. Limitations of small datasets and overfitting risks

A fundamental limitation of this study is the reliance on extremely small datasets, a constraint imposed by the high cost of experimental validation. Operating in this low-data regime introduces an inherent risk of overfitting the Gaussian Process surrogate model to local regions of the formulation space. While the EHVI acquisition function intrinsically

accounts for predictive uncertainty, mitigating premature convergence to false optima, the surrogate model's global accuracy remains constrained by the limited sample size. Consequently, the predictive means generated by the surrogate model should be interpreted as probabilistic guides for navigation rather than high-fidelity physical predictions.

4.3. Scope of applicability and generalizability

The framework was successfully validated on two polymer classes: PU and polyester-based vitrimers. Both systems share relatively continuous formulation-property landscapes. From an algorithmic perspective, the framework operates on a normalized dimensional space and is theoretically chemistry-agnostic. However, its generalizability to highly constrained material systems, such as crystalline materials or conjugated polymers, remains unverified. In crystalline systems, the feasible design space is often a fragmented or highly irregular region dictated by strict stoichiometric constraints. Applying standard Maximin sampling in these contexts might inefficiently assign initial points to physically unrealizable boundary regions. To generalize the framework to such heavily constrained systems, the initialization strategy must be adapted, potentially by employing constrained space-filling designs or incorporating feasibility classifiers into the GP prior before applying EHVI. Therefore, the current framework is suited for combinatorial formulation optimization within a predefined, relatively unconstrained chemical domain.

5. Conclusion

In this work, we established an active learning framework for data-efficient, multi-objective polymer design under severe data scarcity. By systematically evaluating combinations of initialization strategies and acquisition functions across both literature-derived and experimental datasets, we addressed a key challenge in the practical deployment of active learning in polymer research. The results indicate that the combination of Maximin initialization and Expected Hypervolume Improvement provides a robust strategy for reconstructing Pareto fronts. Maximin sampling ensures coverage of boundary regions, reducing extrapolation in the surrogate model, while EHVI selects candidates based on their expected contribution to expanding the Pareto front. Together, this coupling enables efficient exploration of the design space under limited experimental budgets. The framework was experimentally validated using a polyester-based vitrimer system. Within four active learning iterations, formulations were identified that reduced stress relaxation time by 34.5% while increasing Young's modulus by 3.3%. These improvements were obtained from a single physical active-learning campaign rather than averaged over repeated runs. These results demonstrate that meaningful Pareto improvements can be achieved with a small number of targeted experiments, including configurations located in under-explored regions of the design space. This study provides practical guidance for efficient active learning in experimental polymer design under data scarcity. We emphasize, however, that these demonstrations are limited to two systems with relatively continuous formulation-property landscapes, and that generalization to highly constrained or high-dimensional polymer systems requires further validation. The findings suggest that appropriate coupling of initialization and acquisition strategies is critical for achieving high sample efficiency under data scarcity. Future work will focus on integrating the framework with automated experimental platforms to further accelerate materials discovery.

CRedit authorship contribution statement

Yu-Chieh Wei: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Wei-Che Chang:** Methodology, Conceptualization; **Po-An Tsai:** Investigation, Data curation; **Chen-Yu Bai:** Investigation, Data curation; **Ruei-Jing Chu:** Investigation; **Chia-Wen Mu:** Investigation; **Chin-Wen Chen:**

Writing – review & editing, Supervision; **Chuin-Shan Chen**: Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets, implementation codes, and configuration scripts used in this study are publicly available to ensure full reproducibility. They can be accessed via the GitHub repository at: <https://github.com/CM-MAI/Active-Learning-for-Data-Scarce-Multi-Objective-Polymer-Design>.

Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.rineng.2026.111974](https://doi.org/10.1016/j.rineng.2026.111974).

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